Animated version

# Examples of use of MIMOSA, a probabilistic framework for functional data analysis

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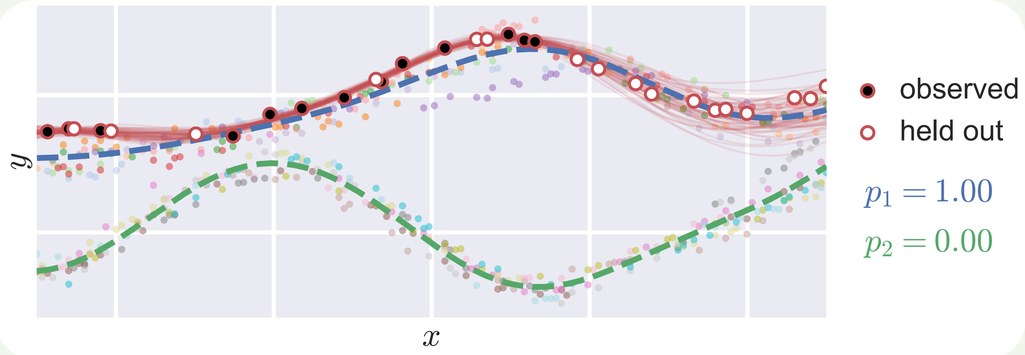
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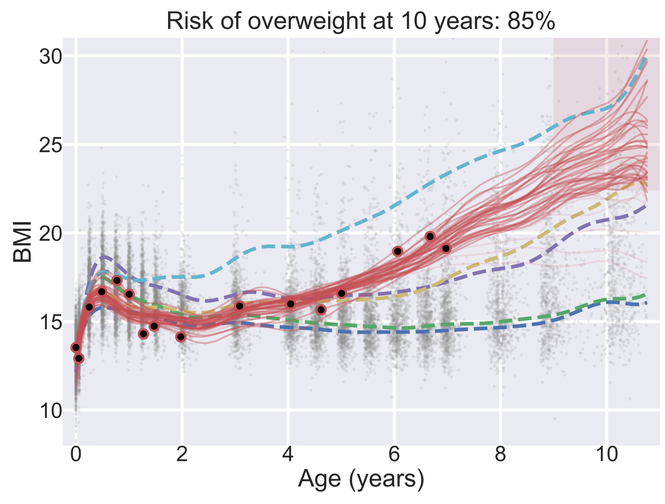
## Intuition: a mixture of multi-task Gaussian processes

| Each sample | = | cluster-specific mean trend | + | sample-specific variation | + | noise |
|---|---|---|---|---|---|---|
| $y_i(x)$ | = | $\Sigma_k Z_{ik} \mu_k(x)$ | + | $f_i(x)$ | + | $\varepsilon_i(x)$ |

uncertainty aware · explainable · scalable · Bayesian · customizable

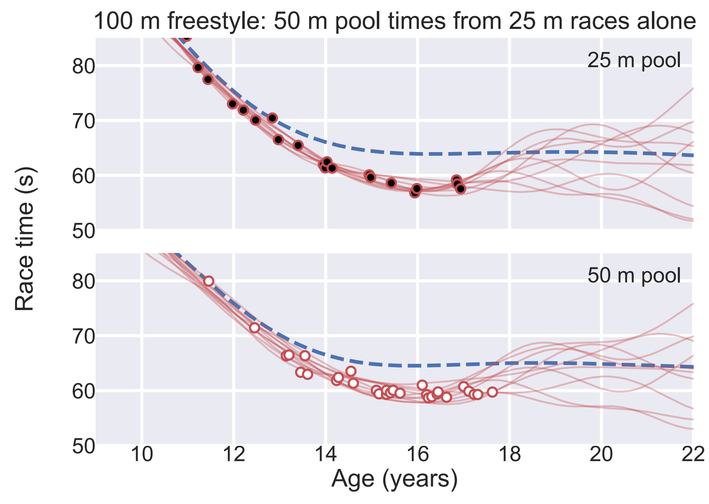



## A · Prospective prediction — child BMI



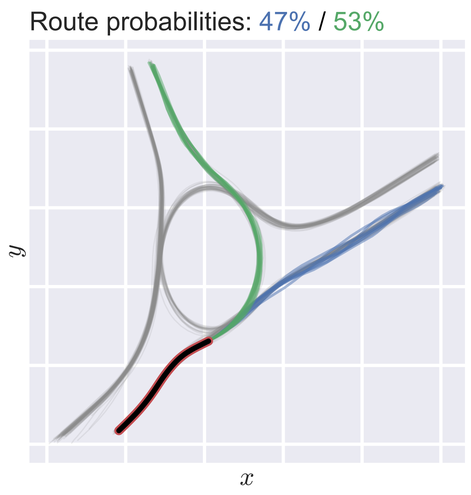

Estimates a child's BMI trajectory and risk of overweight at 10 years from a few early measurements and trends learnt from other children.

## B · Multi-output prediction — swimming



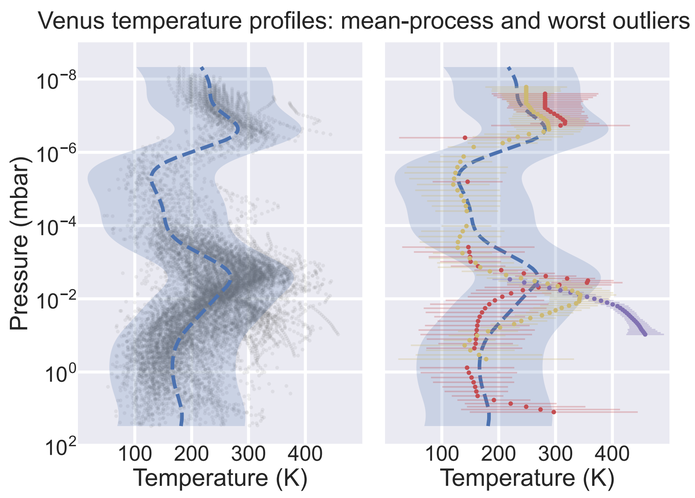

Predicts a swimmer's 50 m pool race times from her 25 m pool races alone, using inter-output correlations learnt from other swimmers.

## C · Trajectory clustering — roundabout




Forecasts a car's trajectory through a roundabout from its first observations, with two likely routes extracted from other cars.

## D · Outlier detection — Venus profiles




Detects outliers among temperature profiles measured on Venus by comparing each sample with its mean process.

**MIMOSA opens up new possibilities in functional data analysis**

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